

School of Psychology
PS3120 – Advanced Cognitive Neuroscience
Cover Sheet

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Word Count: 1791

(Essay only excluding Reference section and this front page cover sheet)

Title: Evaluating the Bayesian brain hypothesis through EEG and fMRI evidence

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Self-reflection on Assignment

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Answer each of the questions to engage in self-reflection and consider your skill development and applications to future contexts.

Self-Reflection on assignment:

- 1) Using feedback from your previous assignments (e.g. PS1104 lit review, PS1108 Executive summary, PS2110 essay, PS3107 essay etc.) describe how you used it to improve this new piece of work.

- 2) Also, using the assessment criteria in the 'Assessment and Feedback' Section on BB, describe how you used it to write this piece of work.

- 3) Is there 1 specific aspect you would like your marker to give extra feedback on? E.g. "I wasn't sure how effective my transition statements were between paragraphs and would appreciate feedback focusing on this aspect".

Insert your essay here

Evaluating the Bayesian brain hypothesis through EEG and fMRI evidence

The brain is increasingly conceptualised as a Bayesian inference machine, continuously generating and refining probabilistic predictions about the sensory world. According to the Bayesian brain hypothesis, perception arises from the integration of prior knowledge with incoming sensory evidence to form optimal interpretations of the environment (Bottemane, 2021). This predictive coding framework enables the brain to operate efficiently by allocating resources primarily to processing unexpected information, known as prediction errors. For instance, Summerfield and Egner (2009) demonstrated that expected or familiar stimuli elicited reduced neural activity in the visual cortex, as measured by fMRI, consistent with the idea that predicted inputs are suppressed. Predictive mechanisms are fundamental to cognitive domains such as attention, perception, and decision-making. While the precise neural implementation of Bayesian inference is still under investigation, ongoing research in this area has significant potential for enhancing our understanding of both typical cognitive function and disorders characterised by impairments in predictive processing.

Building on this foundation, predictive coding models offer a computational account of how the brain might implement Bayesian inference. These models describe a hierarchical system in which higher cortical areas generate top-down predictions that are compared with bottom-up sensory input, prediction errors are used to update internal models (Friston, 2005; Rao & Ballard, 1999). This framework has been applied across cognitive domains such as perception and decision-making. Crucially, predictive coding yields testable hypotheses about the timing and location of prediction error signals in the brain. Cognitive neuroscience tools such as EEG and fMRI are essential for probing these dynamics. EEG offers fine-grained temporal resolution for capturing rapid neural responses to sensory deviations, while fMRI provides detailed spatial maps of the brain regions involved in prediction and error processing. Combined, these methods allow for empirical investigation of the Bayesian brain hypothesis and its role in shaping human cognition.

EEG research allows us to investigate the Bayesian brain hypothesis by identifying expected neural signatures of predictive processing. One key prediction is the presence of mismatch negativity (MMN) which is an event-related potential (ERP) component that should emerge when a sensory input deviates from a predicted pattern. MMN is expected to appear as a negative deflection in the EEG signal in response to deviant stimuli, with greater amplitude for more surprising deviations, reflecting larger prediction errors (Garrido et al., 2009). Another prediction is attenuation of early ERP components such as P1 and N1 when stimuli match expectations, this is consistent with the idea of predictive suppression. EEG can also be used to examine ongoing neural oscillations which can offer deeper insights into the temporal dynamics of predictive processing. Neural oscillations are rhythmic patterns of electrical activity produced by neurons; the Bayesian framework predicts that alpha and beta oscillations are involved in encoding and maintaining prior expectations (Arnal & Giraud, 2012), while gamma-band activity tracks feedforward transmission of prediction errors (Bastos et al., 2015). These predicted EEG signatures provide measurable physiological correlates of the computational processes by the Bayesian brain model.

fMRI research enables us to test predictions of the Bayesian brain hypothesis by identifying spatial patterns of neural activity associated with expectation, prediction errors, and hierarchical inference. It provides valuable insights into the neural correlates of predictive processing by measuring blood-oxygen-level-dependent (BOLD) signals as an indirect index of neural activity. As a brain area becomes more active, oxygen consumption increases therefore more oxygenated blood flows to the active area; these BOLD signals measure metabolic activity in the brain indicating when and where there is an increase in the workload. Bayesian frameworks have been central in modelling the brain as a probabilistic learner in decision-making contexts, with fMRI data supporting theoretical accounts of how the brain encodes uncertainty and updates beliefs. fMRI studies have shown greater activity in primary auditory or visual cortices when sensory input violates prior expectations, consistent with the idea that prediction errors are passed forward to update internal models (den Ouden et al., 2012). Behrens et al. (2007) took BOLD data from specific regions of interest and found higher volatility and variance when circumstances changed, indicating increased BOLD activity is a result of unexpected sensory input and therefore learned information.

While EEG offers strong temporal resolution for identifying prediction-related signatures, its strengths and limitations become clearer when examining experimental designs that probe the Bayesian brain hypothesis more deeply. Chennu et al. (2013) used a roving oddball paradigm in which participants listened to sequences of sounds with varying levels of regularity and complexity. They found that both MMN and late ERP components such as the P300 scaled with the hierarchical depth of auditory violations, showing EEG's capacity to capture layered prediction errors in real time. The dependent variable here was ERP amplitude across conditions reflecting different probabilistic expectations. The findings showed that well-known MMN and P300 components reflect a hierarchy of predictive processing, analysis linked these components to distinct temporal and frontal brain regions. The EEG data provided an integrated view of ERPs within the framework of hierarchical predictive coding resulting in a bridging of empirical findings in Bayesian theories.

However, EEG's low spatial resolution means that while we can observe when prediction errors occur, we cannot precisely determine where in the brain they are generated. This limits our ability to map the full hierarchical structure posited by predictive coding theories. Greater spatial resolution allows for researchers to discriminate between different brain regions with greater accuracy, as electrical activity is often detected in several brain regions simultaneously, it makes it difficult to draw accurate conclusions in empirical research (*Biopsychology – Studying the Brain – Evaluating EEG & ERP, n.d.*). EEG's limitation of determining an exact location and activity of neural sources, known as the inverse problem (Michel & Brunet, 2019), is ill-posed because a number of different configurations can report the same electrical scalp pattern back to the electrodes, this makes it impossible to posit a final conclusion without additional constraints.

While fMRI lacks the temporal precision of EEG, its high spatial resolution makes it a powerful tool for mapping the anatomical distribution of prediction-related processes across the brain. For example, den Ouden et al. (2008) used fMRI to show that BOLD responses in primary visual cortex increased when an expected visual stimulus was violated, this response grew stronger as participants learned the predictive relationship, indicating a learning-based modulation of sensory signals. These findings support the predictive coding framework, suggesting that fMRI can reveal how top-down expectations shape sensory processing. Summerfield and de Lange (2014) reviewed numerous fMRI studies demonstrating that predicted stimuli evoke reduced BOLD responses compared

to unpredicted ones, a phenomenon known as expectation suppression. Ozel et al. (2023) discussed how fMRI is non-invasive, highlighting its advantages over techniques like EEG, which require the placement of electrodes on the scalp. The ability to detect changes in blood oxygenation without physically penetrating the body makes it a safe and comfortable technique for participants which is advantageous in clinical settings.

However, fMRI's reliance on hemodynamic responses means it cannot capture the rapid, millisecond-scale updates that characterise prediction error signalling in real time. FMRI's BOLD signal has low temporal resolution because it reflects slow changes in blood rather than fast neural activity (Glover, 2011). Additionally, the BOLD signal is an indirect proxy for neural activity, influenced by vascular and metabolic factors, which complicates interpretation. While fMRI excels at identifying where in the brain predictions and errors occur, it cannot clarify the precise timing or causal sequencing of these computations. As such, fMRI provides a spatially detailed but temporally coarse view of the hierarchical architecture underlying Bayesian inference in the brain.

While EEG and fMRI differ significantly in their temporal and spatial characteristics, they both create data that reflects the same neuronal processes, the complementary strengths of both techniques become apparent when employing them both simultaneously. Van Atteveldt et al. (2015) found that participants detected target sounds more quickly and accurately when they occurred in rhythmically structured streams compared to randomly timed ones. FMRI revealed greater activation in the auditory cortex during the processing of random sequences, this may show increased metabolic demand due to less efficient processing, on the other hand, EEG data provided finer temporal insights, showing larger amplitudes and delayed latencies for irregular stimuli. These findings together suggest that rhythmic predictability enables more efficient neural entrainment; oscillatory activity also aligned to expected input phases to optimise processing, indicating that the brain times its activity to respond when important stimuli is most likely to appear to make processing more efficient. This shows that fMRI and EEG offer distinct yet converging perspectives, fMRI captures where processing efficiency varies while EEG shows time-series data of how these efficiencies unfold over time. This synergy underscores the value of integrating both modalities to understand dynamic brain function in contextually structured environments.

FMRI enables a detailed evaluation of how psychological processes such as expectation and prediction error are instantiated across distributed neural circuits. its ability to track changes in inter-regional connectivity has been instrumental in showing how top-down predictions and bottom-up sensory signals interact hierarchically. Through techniques like dynamic causal modelling, Friston et al. (2003) used a Bayesian framework to estimate effective connectivity from fMRI data, allowing researchers to interpret fMRI experiments to reveal how contextual factors influence brain network integration. Iglesias et al. (2013) demonstrated that the combination of computational modelling and fMRI has enabled researchers to map hierarchical prediction errors onto distinct brain regions, supporting Bayesian models of cognition. The findings highlight fMRI's unique ability to abstract computational processes to distributed neural substrates in humans.

The high temporal resolution of EEG allows it to effectively contribute to understanding the Bayesian brain hypothesis by revealing the temporal dynamics of prediction errors. Nieuwenhuis et al. (2004) used EEG to identify N2 and P3 components associated with error detection and prediction updates. The sensitivity of these components to different aspects of feedback highlights EEG's strengths in distinguishing between utilitarian and performance-based reactions in real-time. This showcases EEG's unique capacity to link

temporal brain dynamics to cognitive processes such as error detection and outcome evaluation, offering valuable insights into the brain's predictive mechanisms.

In conclusion, EEG and fMRI each provide valuable but distinct insights into the neural mechanisms underlying predictive processing. EEG offers high temporal resolution, making it well-suited to capture the timing of prediction errors, whereas fMRI offers high spatial resolution enabling precise localisation of the brain regions involved. For a comprehensive understanding of the Bayesian brain hypothesis, combining both techniques is recommended. This underscores the importance of multimodal neuroimaging in cognitive neuroscience research and in developing more robust models of brain function. Integrating these methods not only compensates for their individual limitations but also enables a more accurate and comprehensive understanding of predictive processing, a deeper comprehension of the brain's predictive architecture is critical for scientific advancements.

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